



ADCA + HARM

**SAFE MIDDLEWARE & REAL-TIME SAFETY SHIELD
FOR VISION-LANGUAGE MODEL ROBOTICS**

**Presented by
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PROBLEM STATEMENT:

UNFILTERED VLM's IN PHYSICAL ROBOTICS

Vision-Language Models (VLMs) are powerful reasoners, but inherent probabilistic failures make direct hardware execution unsafe.



**Spatial Hallucinations &
Out-of-Bounds Targets**



**Schema Corruptions &
Parsing Failures**

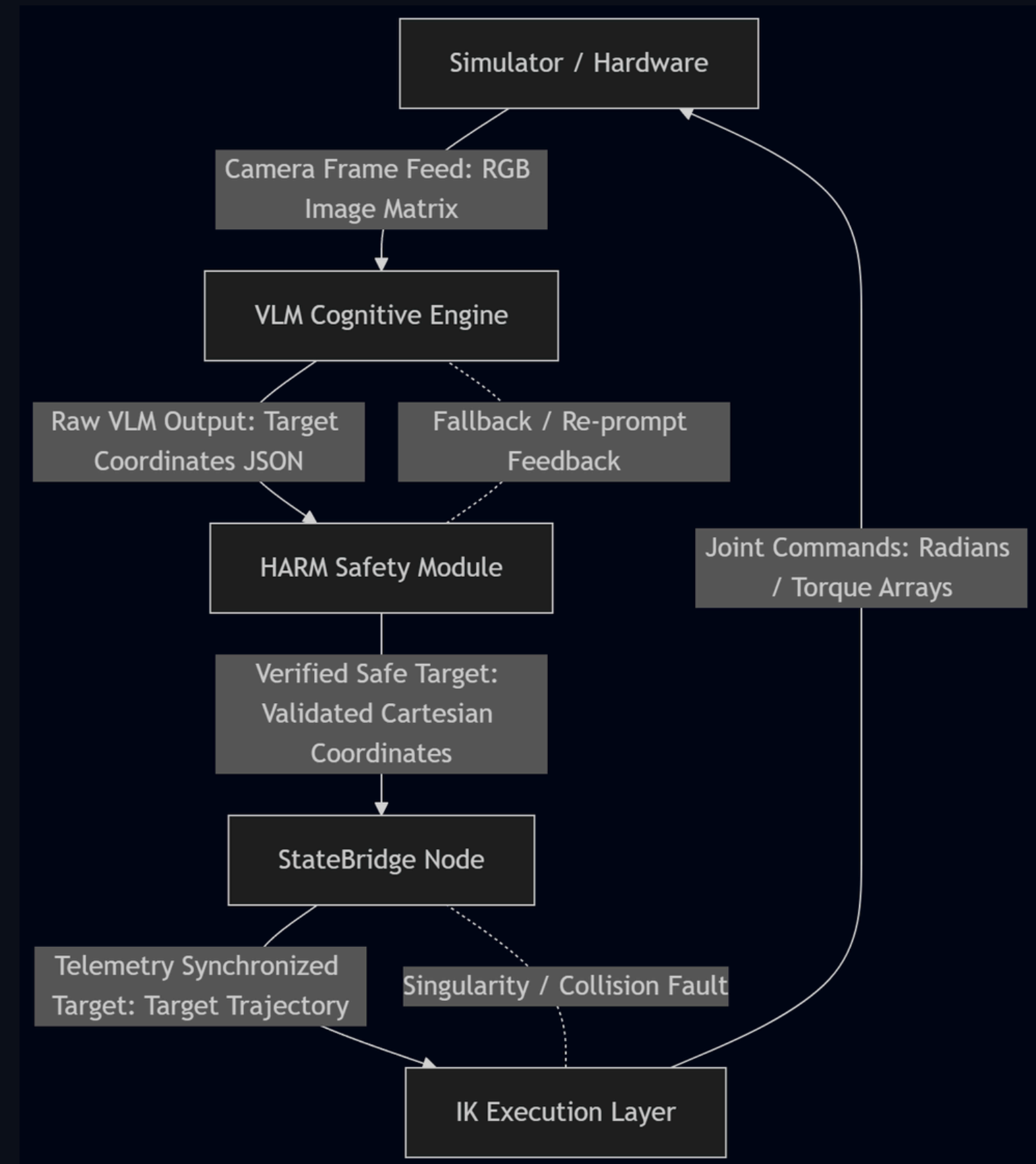


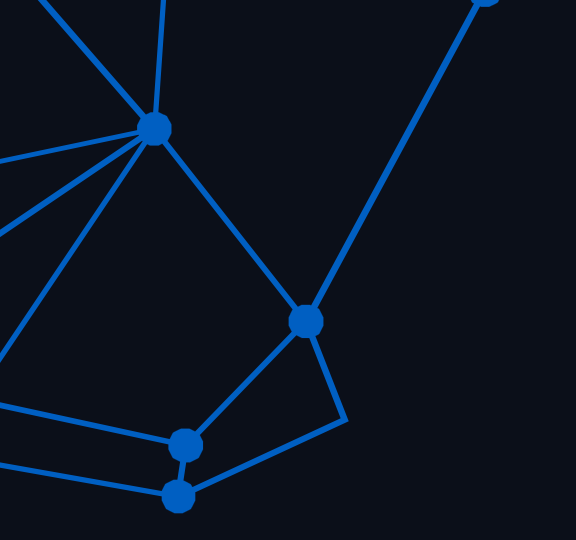
**Kinematic Speed
& Latency Spikes**

SYSTEM ARCHITECTURE: PIPELINE FLOW

Decoupling high-level AI reasoning
from real-time execution via
deterministic safety layers.

- ✓ **Deterministic Interception**
- ✓ **Async Queue Management**
- ✓ **Closed-Loop Feedback**





HARM: 3-STAGE SAFETY VALIDATION

Multi-layered defense-in-depth architecture filtering AI commands before motor execution.

STAGE 1: SCHEMA & FORMAT

Pydantic Structural Guard

Key Features:

- Strips markdown fences (`` ` ` ` json).
- Validates action types (MOVE, HOLD, STOP).
- \$N=1\$ Re-prompt Loop: Sends failure details back to VLM for immediate single-attempt self-correction.

STAGE 2: WORKSPACE BOUNDS

3D Reachability Shell

Key Features:

- Enforces dual-sphere spatial limits $R_{\min} = 0.2\text{m}$, $R_{\max} = 1.0\text{m}$
 - 3D Target Clipping: Projects out-of-bounds targets back onto the safe reachable surface shell.
 - Prevents joint hyperextension and collisions.
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HARM: 3-STAGE SAFETY VALIDATION

Multi-layered defense-in-depth architecture filtering AI commands before motor execution.

STAGE 3: KINEMATICS & SPEED

Velocity Scaling & Fallback

Key Features:

- Enforces joint velocity ceiling (≤ 0.8 rad/s)
 - Dynamic Time Scaling: Scales execution time to smooth aggressive joint acceleration.
 - Safe Hold Fallback: Triggers zero-velocity hold if unresolvable faults occur.
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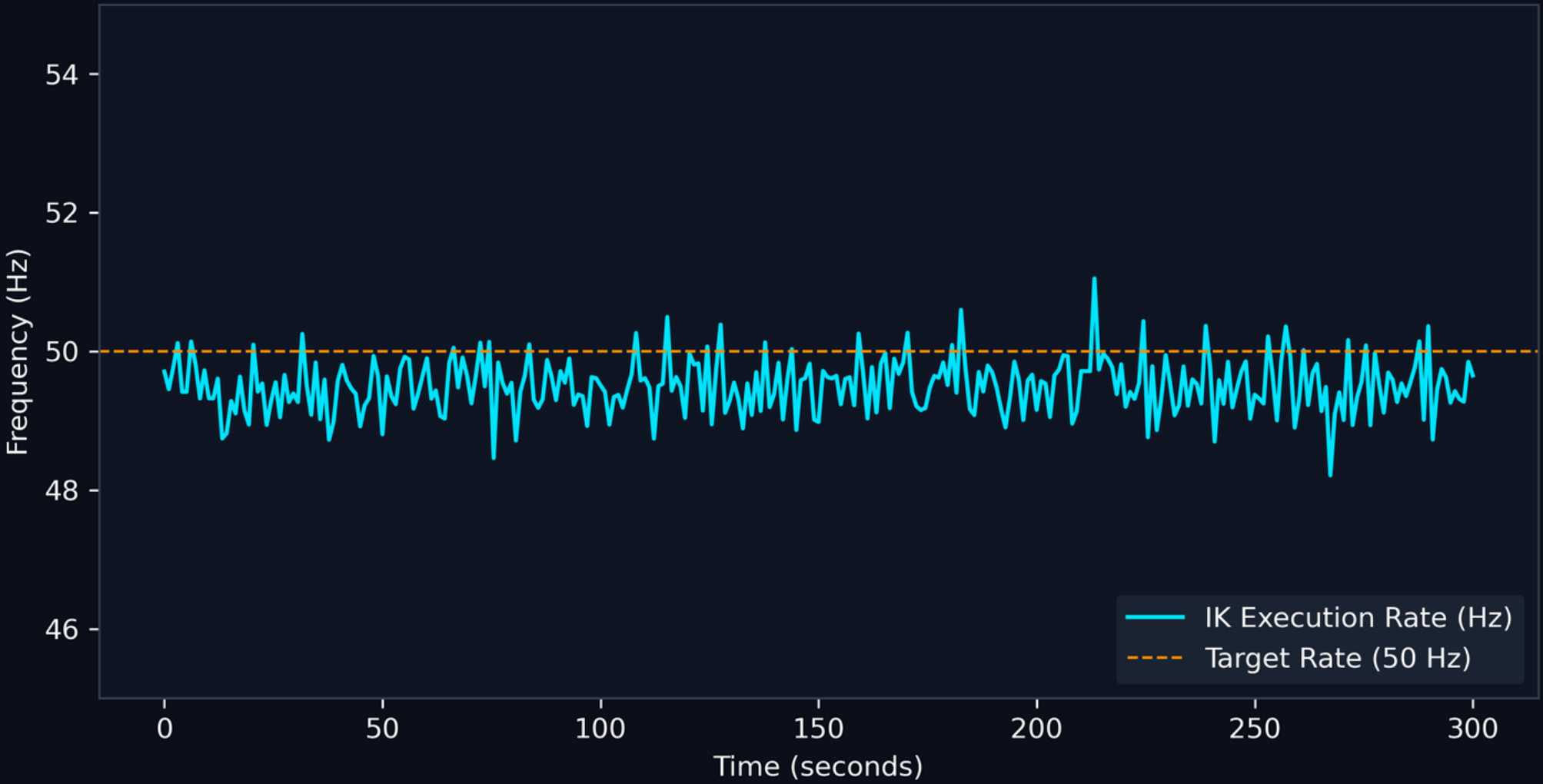
LIVE DEMO: REAL-TIME SAFETY INTERCEPTION

Executing live out-of-bounds VLM payload injection via Construct
RDS environment.

REAL-WORLD HARM CATCH RATE & METRICS

Quantitative evaluation of 50 real-world VLM calls across E1, E2, and E3 scenarios.

IK Controller Frequency Timeline (5-Min Stress Test)



IK FREQUENCY
TIMELINE PLOT

HARM CATCH RATE TABLE

Metric Category	Count / Value	Percentage
Clean Passed	35 / 50	70%
Stage 1 (Schema Intercept)	7 / 50	14%
Stage 2/3 (Spatial Recovery)	5 / 50	10%
Safety Fallbacks Triggered	3 / 50	6%
Total Catch Rate	15 / 50	30%

BENCHMARK SPEEDUP FACTOR

Key Metrics:

- HARM Intercept Latency: < 5 ms
(Avg: ~ 1.2 ms)
- Cloud VLM Re-query Latency:
1500 ms – 3000 ms

Highlight Callout:

- 300x Speedup Factor

Local deterministic safety resolution
vs. re-prompting heavy cloud VLMs

LIMITATIONS & EDGE CASES

Structural constraints, architectural trade-offs, and engineering mitigation strategies.

WORKSPACE GEOMETRY

Title: SPHERICAL SHELL APPROXIMATION

Constraint:

- Stage 2 uses a simplified radial shell $R_{\min} = 0.2\text{m}$, $R_{\max} = 1.0\text{m}$. It does not model complex non-spherical manipulator geometry or tabletop obstacles.

Mitigation / Roadmap:

- Upgrade Euclidean radial checks to mesh-based octree representations or dynamic URDF collision checking.

RE-PROMPTING BUDGET

Title: SINGLE RE-PROMPT RETRY LIMIT

Constraint:

- Stage 1 enforces a strict single re-prompt budget ($N=1$) for malformed VLM JSON to prevent latency accumulation in control loops.

Mitigation / Roadmap:

- Fine-tune base system prompts or adopt guided JSON generation frameworks (e.g., Outlines / Guidance).

LIMITATIONS & EDGE CASES

EXECUTION RUNTIME

Title: ASYNCIO EVENT LOOP CONSTRAINTS

Constraint:

- Running explicit ROS 2 executor loops inside existing asyncio loops (e.g., Jupyter/Colab) can cause thread-blocking synchronization issues.

Mitigation / Roadmap: Execute:

- ROS 2 callbacks in `MultiThreadedExecutor` or isolate middleware nodes in dedicated subprocess workers.



Production Safety Invariant:
HARM acts strictly as a high-level supervisory filter. It does not replace hardware-level emergency stop (E-stop) circuits or joint-level motor drivers.

WHAT'S NEXT

Scaling deterministic safety middleware for next-generation physical robotics.

FUTURE ROADMAP

Engineering Roadmap:

- Mesh-Based Octree Collisions: Replace radial Euclidean shells with real-time 3D spatial octree grids for complex tabletop environments.
- Multi-Threaded ROS 2 Executors: Transition HRAFMiddlewareNode callbacks to isolated subprocess workers for zero-thread-blocking latency.
- Guided Schema Generation: Integrate Outlines/Guidance for 100\% guaranteed structural JSON output at token generation.



WHAT'S NEXT

PRODUCTION INVARIANT & TAKEAWAY

Production Safety Invariant:

- HARM sits strictly as a high-level supervisory filter between cognitive VLMs and low-level motion controllers.
- It complements—rather than replaces—hardware E-stops and motor torque limits.

Key Takeaway:

- Unfiltered VLMs are unsafe for direct hardware execution. HARM provides low-latency (<5 ms), deterministic protection for real-time robotic systems.





THANK YOU

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